



Water Quality Anomaly Detection & Health Risk Predictor

A Full Stack AI Platform with ML-Based Anomaly
Detection and AI-Powered Health Risk Analysis

*-By
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The Human Cost of Contaminated Water

Global Impact (WHO data)

2 Billion

People consuming water contaminated with biological waste.

485,000

Preventable deaths annually from diarrhoeal diseases (cholera, typhoid, dysentery)

Local Crisis in India (NITI Aayog data)

37.7 Million

People affected by waterborne diseases annually.



Vulnerability: Children under five face the highest mortality risk, alongside severe long-term neurological risks from heavy metals like lead and arsenic.

A Reactive, Delayed, and Manual Paradigm



Contamination
Occurs

The Critical Gap: Exposure Window

By the time laboratory results return and authorities respond,
large populations have already been exposed.



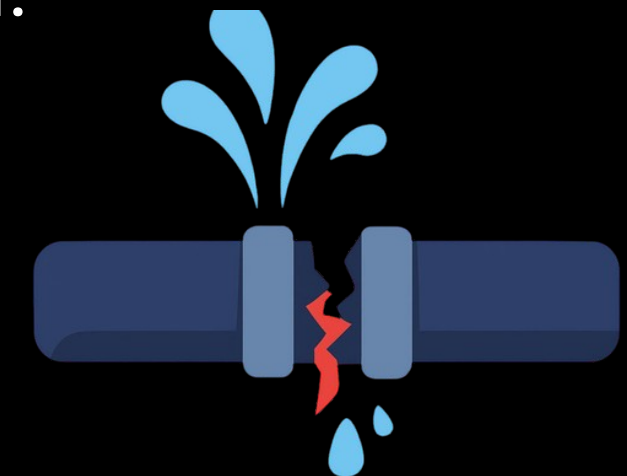
Lab Report
Arrives

Key Failures

Infrequent: Periodic sampling leaves long blind spots.



Slow: Results take days or weeks to process.



Reactive: Responses only occur after an outbreak has begun, rather than preventing it.

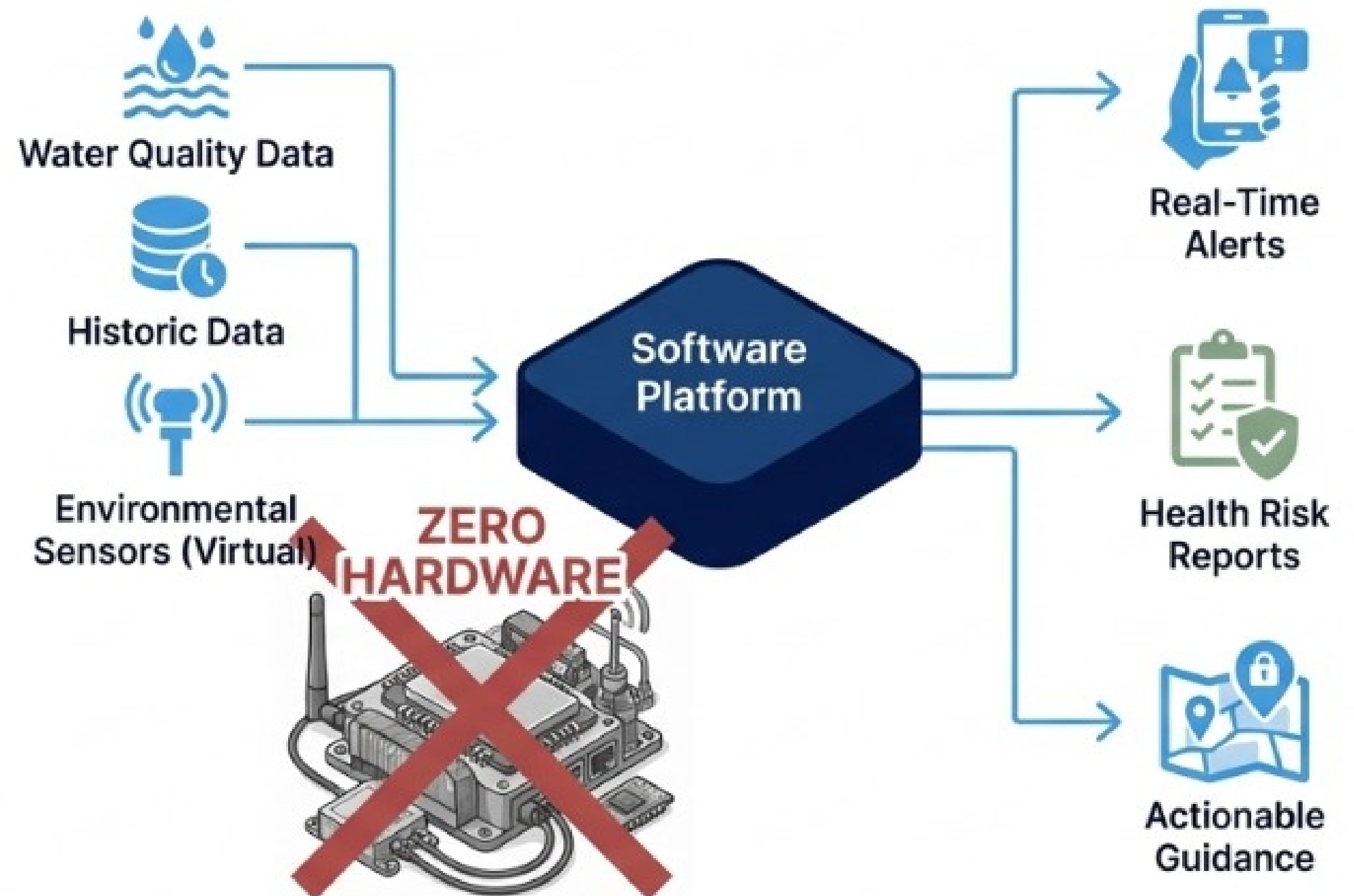
An Intelligent, Software-Only Solution

Core Concept:

A full-stack web platform combining ML anomaly detection and AI health risk analysis to democratise water quality monitoring.

The Differentiator:

Achieves real-time monitoring, intelligent anomaly detection, and actionable guidance entirely through software—eliminating the dependency on physical IoT infrastructure and making it accessible to resource-constrained environments.



Core System Objectives

Five core capabilities that enable intelligent water quality monitoring

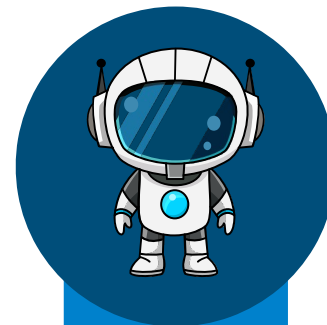
Obj 1: Multi-Source Ingestion

Manual, CSV, and automated API synchronisation.



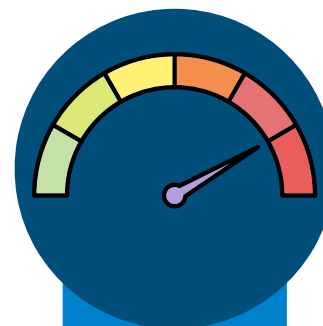
Obj 2: ML Detection

Isolation Forest & Z-score analysis across 8 parameters.



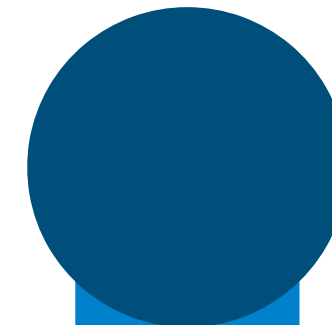
Obj 3: Risk Scoring

0–100 composite scoring based on WHO and BIS thresholds.



Obj 4: AI Integration

Claude API for human-readable health risk assessments.



Obj 5: Real-Time Dashboard

Visualisations, anomaly feeds, and automated alerts.



A Modern, Scalable Technology Stack

01

Frontend

React 18, Vite, Tailwind CSS, Recharts.
Fast, component-based visualisation

04

ML Microservice

Python 3.11, FastAPI, scikit-learn
Async type-safe endpoints

02

Backend API

Node.js 20, Express 4, node-cron
Lightweight orchestration

05

AI Analysis

Anthropic Claude API
Superior contextual health domain reasoning

03

Database

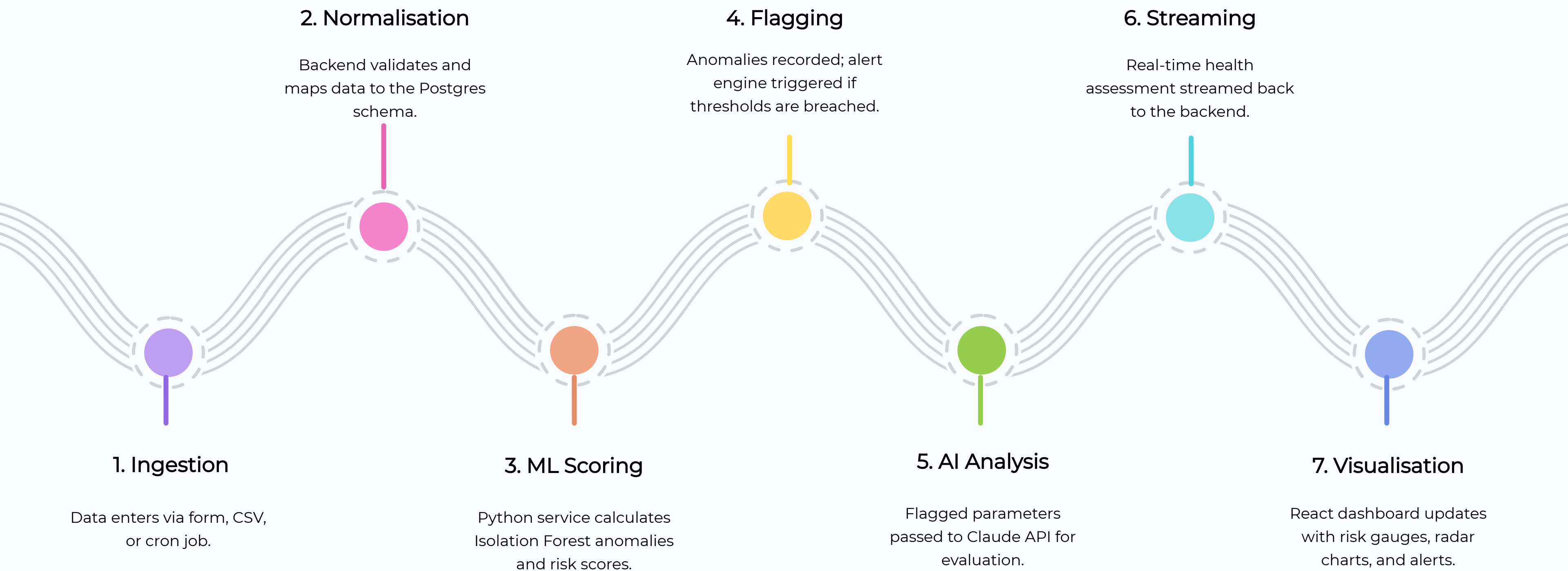
PostgreSQL 15
ACID compliance, strong typing for scientific data

06

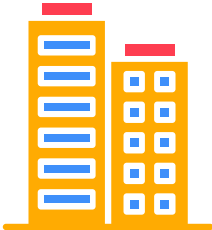
Security

JWT, bcrypt
Stateless, scalable authentication

End-to-End Data Flow



DELIVERING BROAD STAKEHOLDER VALUE

	Stakeholder	Current Challenge	Proposed Benefit
	• Urban Households	No visibility into tap safety.	• Instant risk scoring and health guidance.
	• Rural Communities	No lab infrastructure.	• Free software-based monitoring.
	• Municipal Authorities	Manual, delayed reporting.	• Automated alerts and trend analysis.
	• Health Departments	Reactive outbreak response.	• Predictive early-warning systems.



THANK YOU